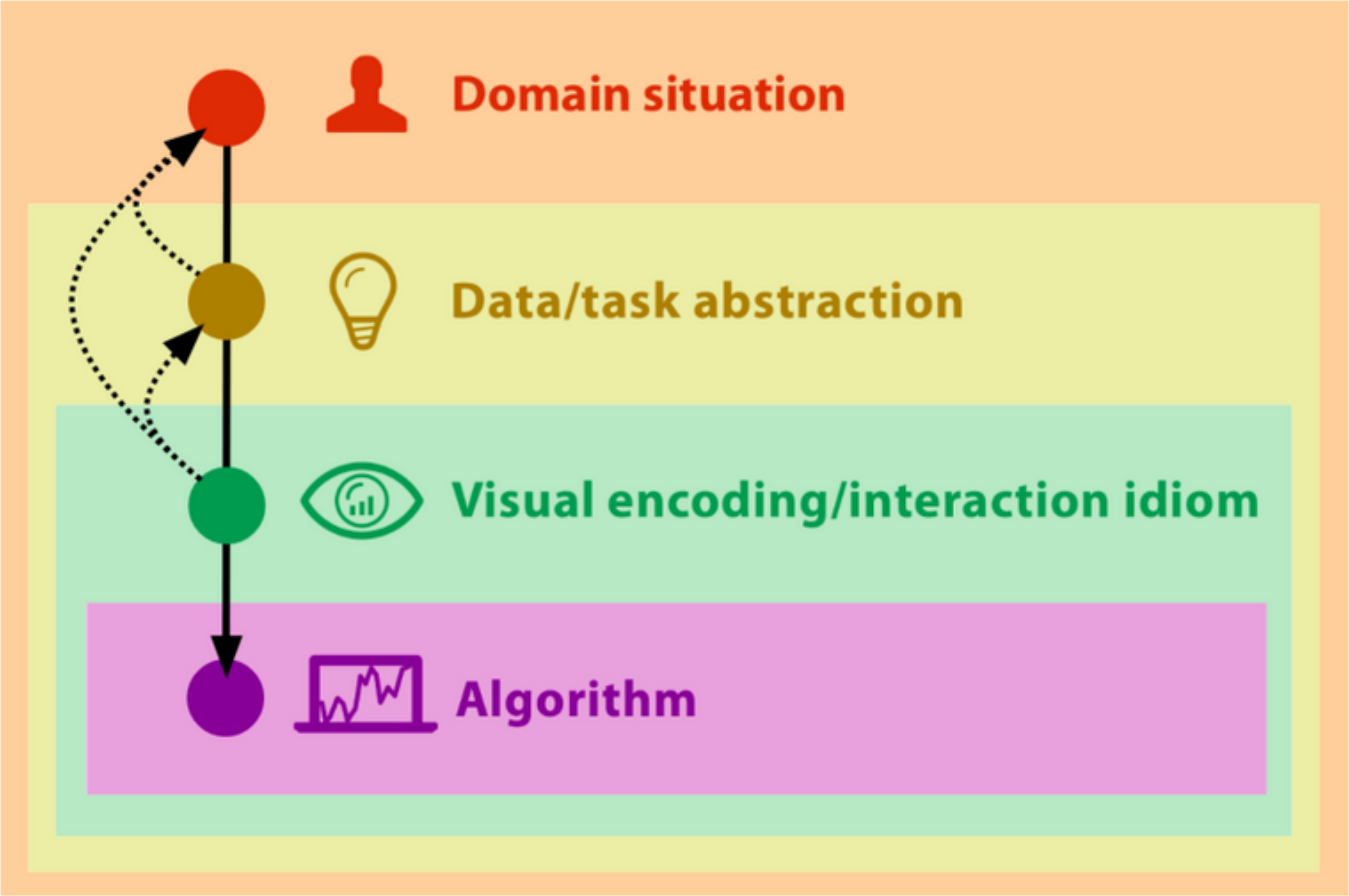
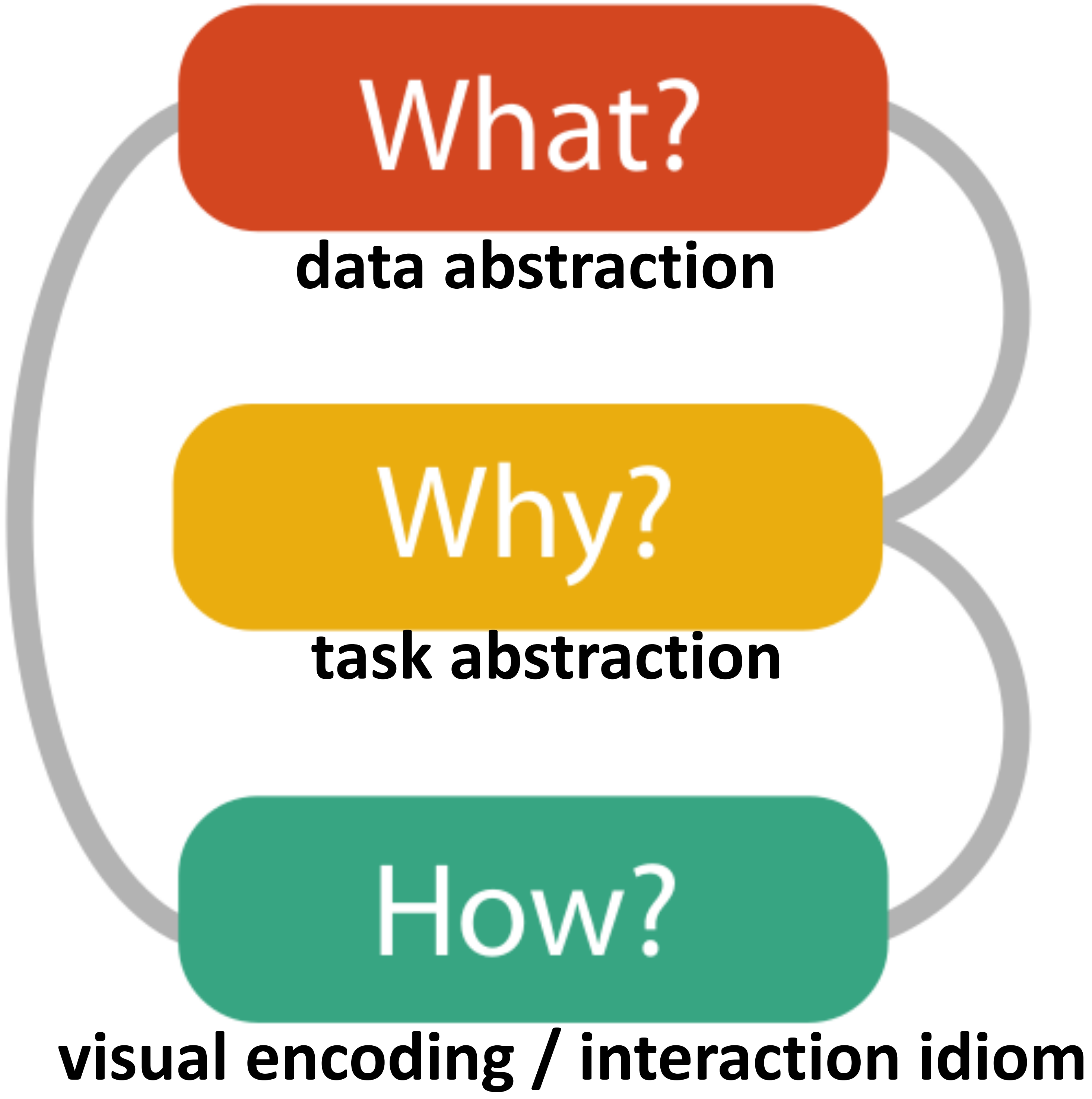
Visualization Guidelines

DSK808 Data Visualization

Fall 2025

Participatory Visualization Design





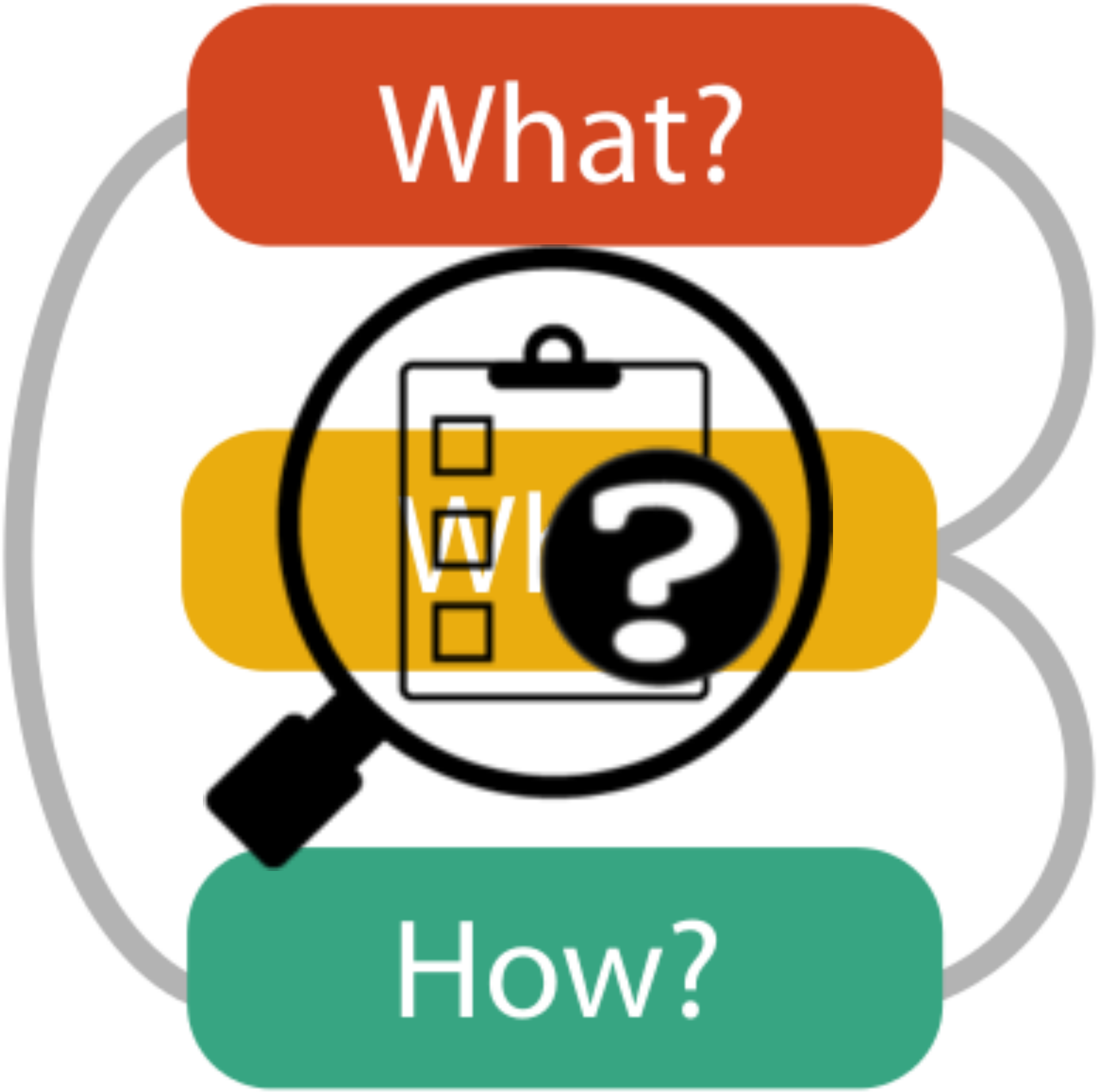
Participatory Visualization Design

Attributes
ordinal
numerical
categorial
relational

Availability
static
dynamic

Actions
analyze
present
search
derive
...

Visual Representations
tag clouds
maps, charts
timelines
...



Data types
entities
relations
attributes

Data set types
2D tables
nD tables
networks

Targets
trends
outlier
features
distributions
correlations
...

Interactions
select
filter
navigate
...

Information Measure and Transfer

- How to **measure** the information presented in a visualization?

- **Expressiveness** measures the concentration of information
  - Expressive if all and only the existing information is present
  - Ratio $M_{\mathrm{exp}} \in [0,1]$ of actually displayed information to information intended to be presented
    - optimal if $M_{\mathrm{exp}} = 1$
- **Effectiveness** measures the cost of information perception
  - Ratio $M_{\mathrm{eff}} \in [0,1]$ taking into account effort of interpretation as well as cost of graphical rendering
  - $M_{\mathrm{eff}} = 1 / (1 + \mathrm{time}_{\mathrm{interpret}} + \mathrm{time}_{\mathrm{render}})$

Motivation

What is the best chart to visualize my KPIs?

Visualization Guidelines

"A guideline embodies wisdom advising a sound practice. This may be a course of action to take or to avoid in achieving a goal."

"Good visualizations should maximize data-ink ratio".

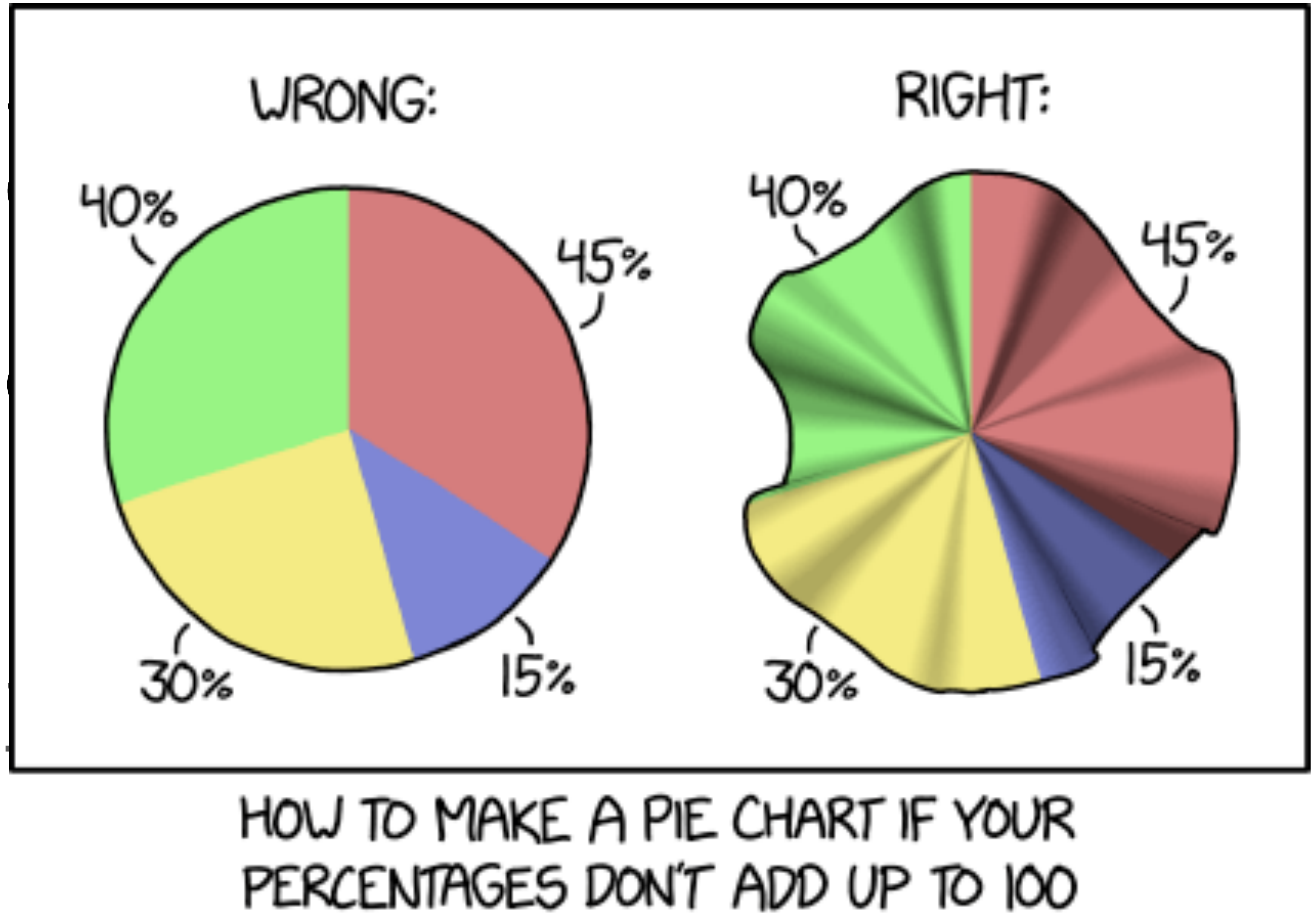
Don't make users do "visual math".

"Overview first, zoom and filter, details on demand".

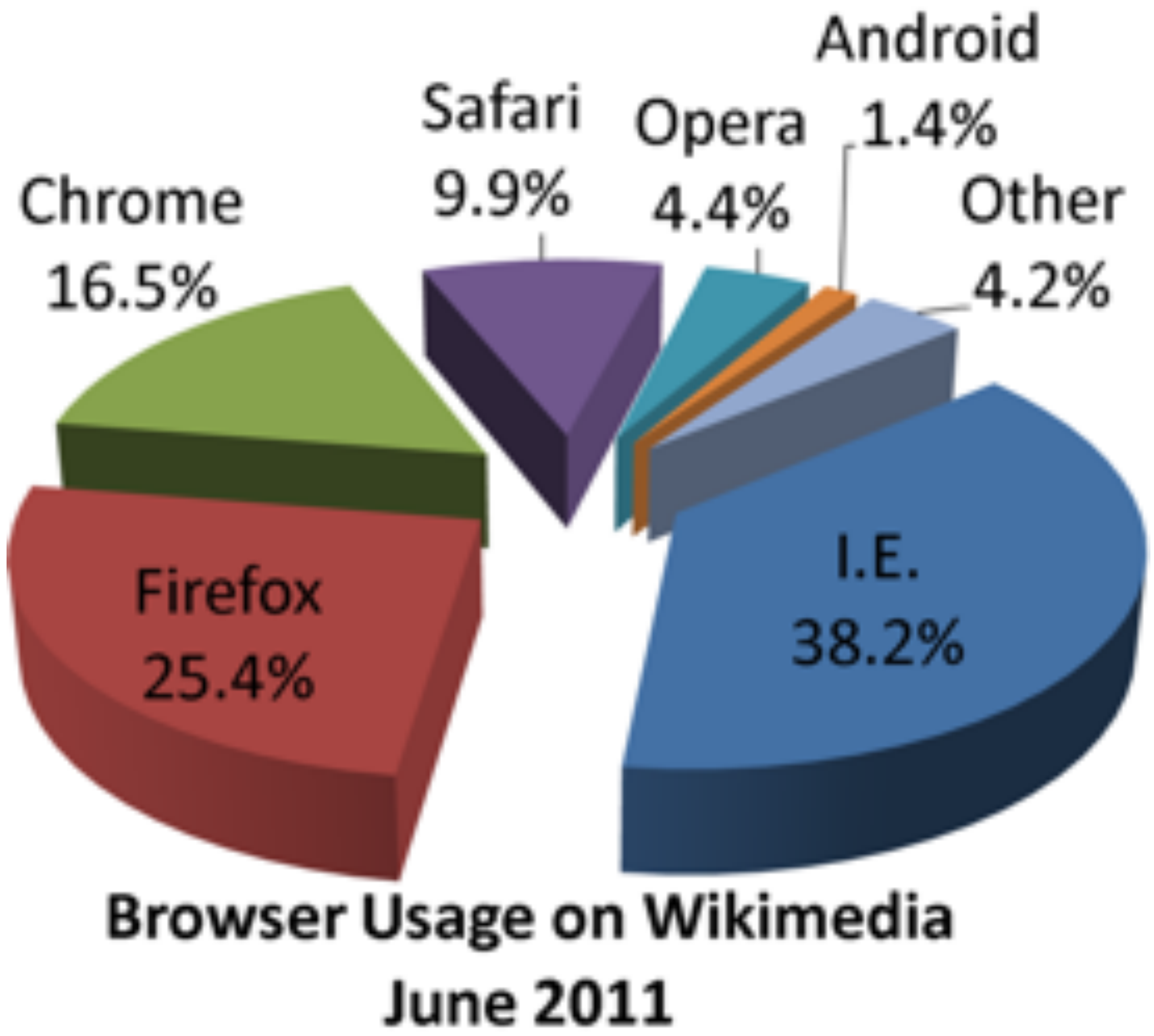
Chen, M., Grinstein, G., Johnson, C. R., Kennedy, J., & Tory, M. (2017). Pathways for theoretical advances in visualization. *IEEE computer graphics and applications*, *37*(4), 103-112.

Pie Charts!

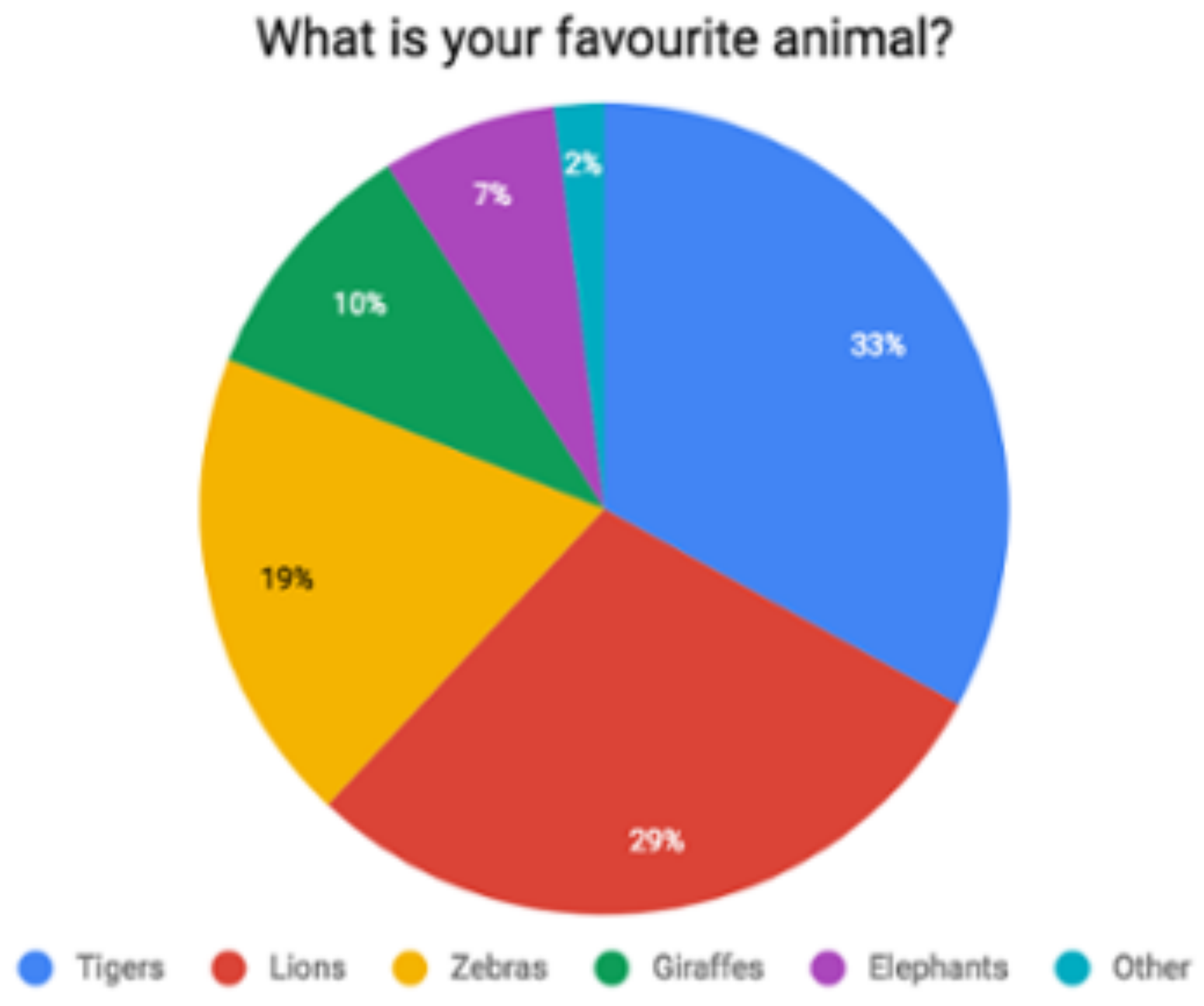
Bertin’s Visual Variables

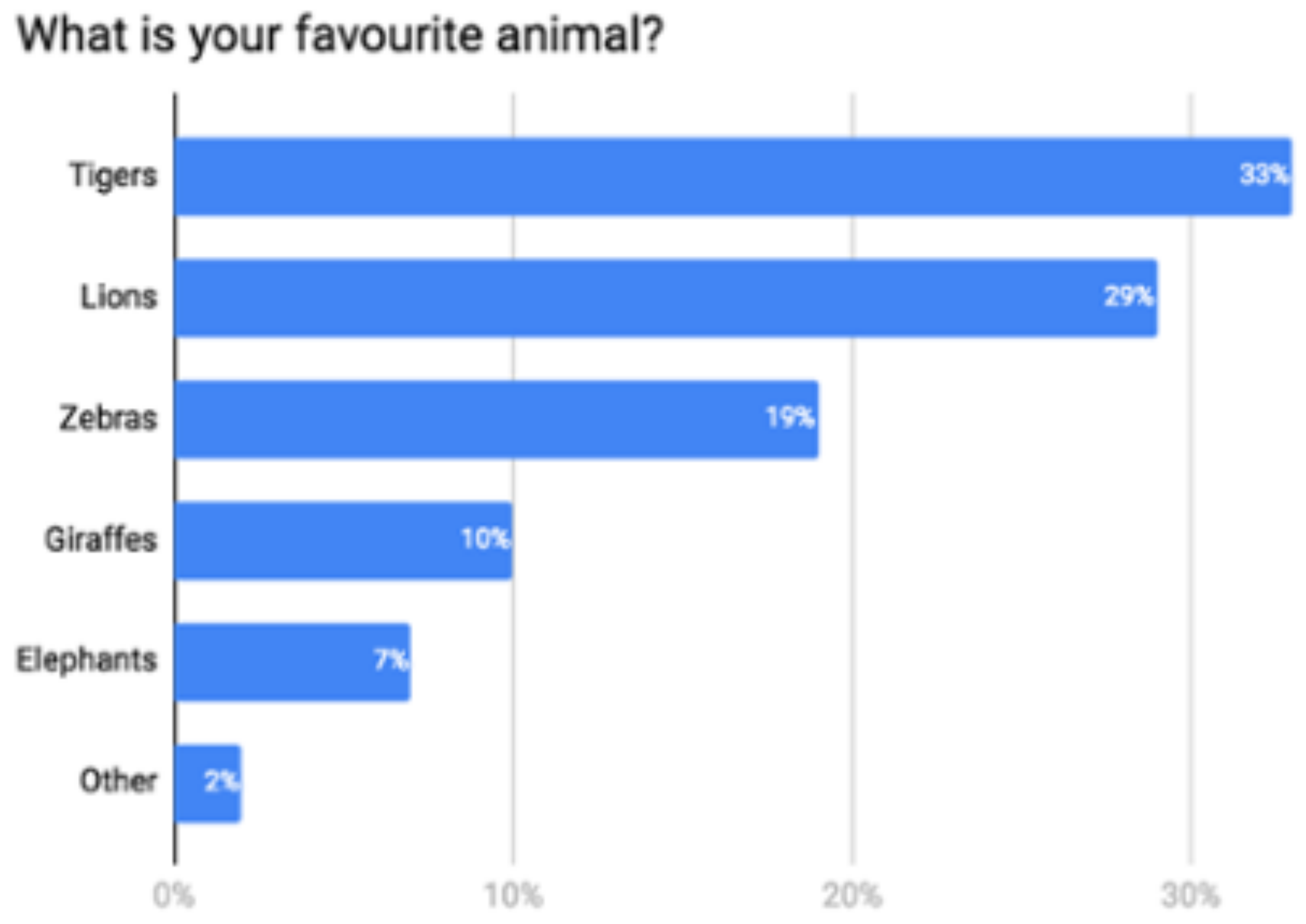


Should I use 3D?



Why you should't use pie charts - Tips for better data visualisation



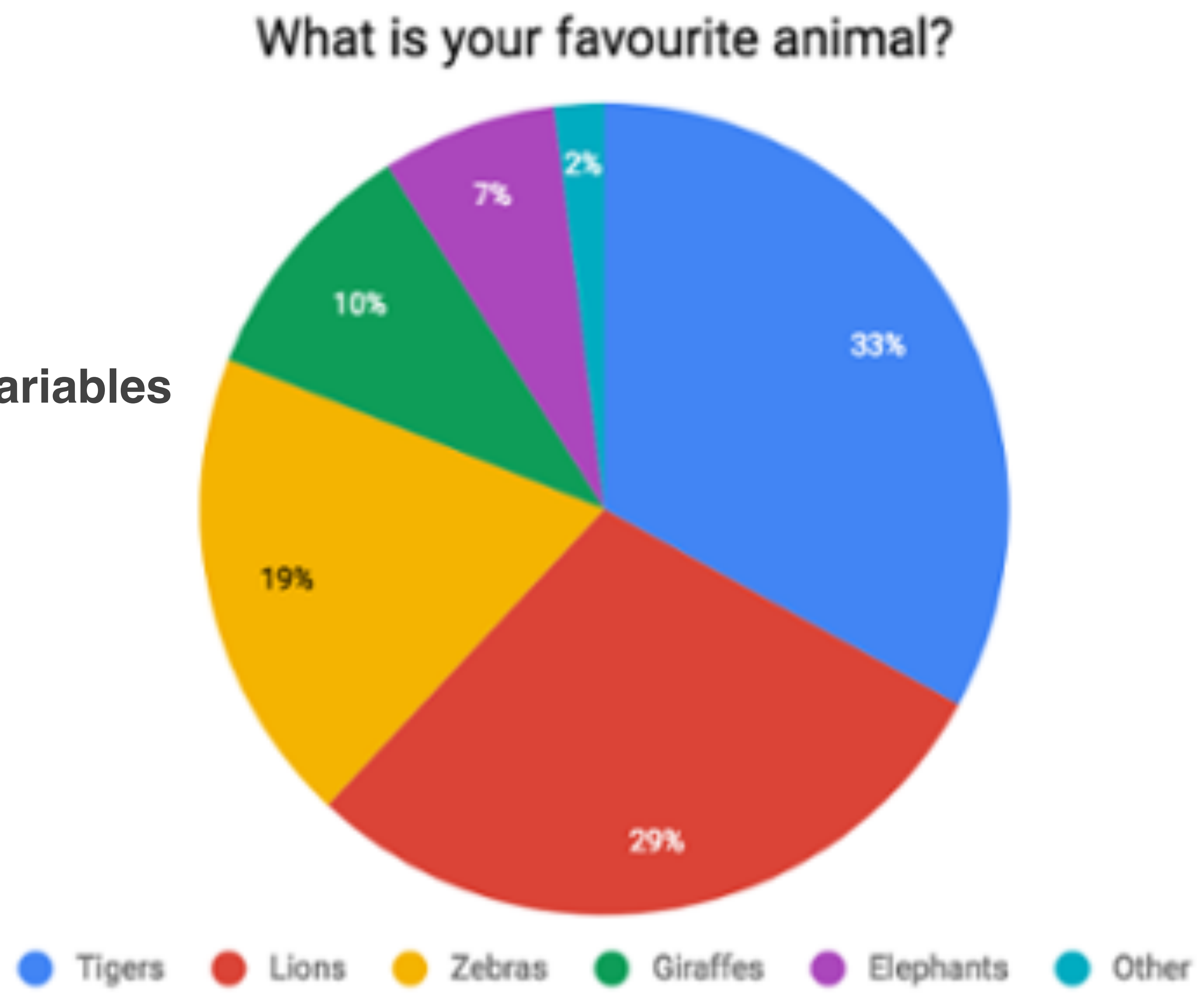


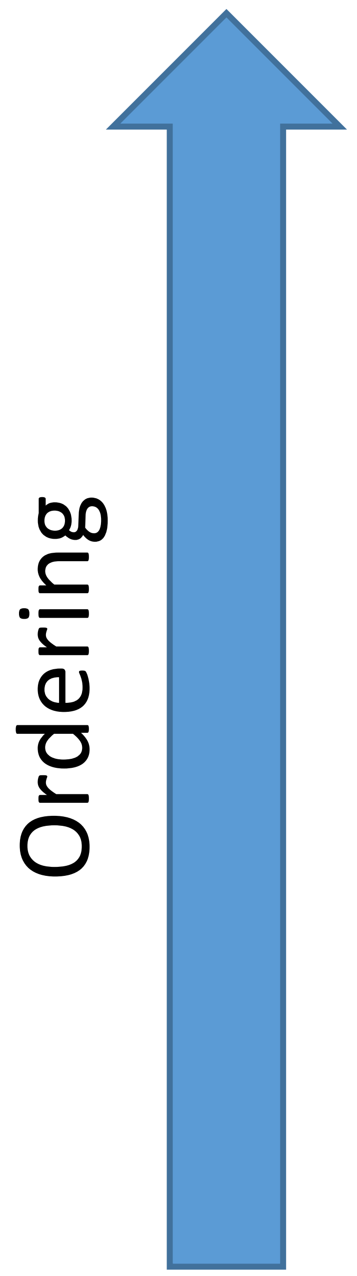
Eagereyes: https://eagereyes.org/techniques/pie-charts

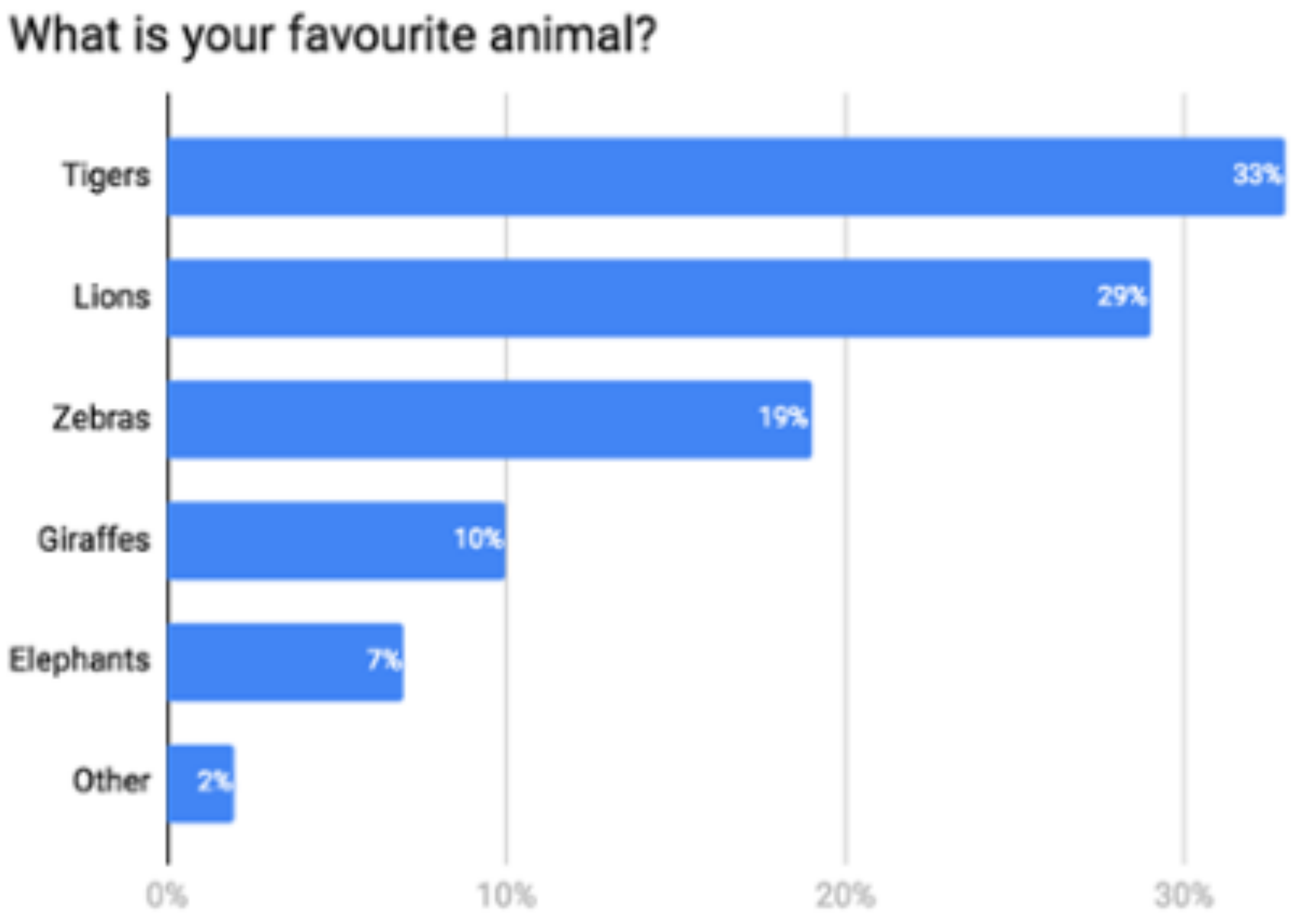
Why you should't use pie charts - Tips for better data visualisation

Bertin's Visual Variables

-Position
-Size
-Shape



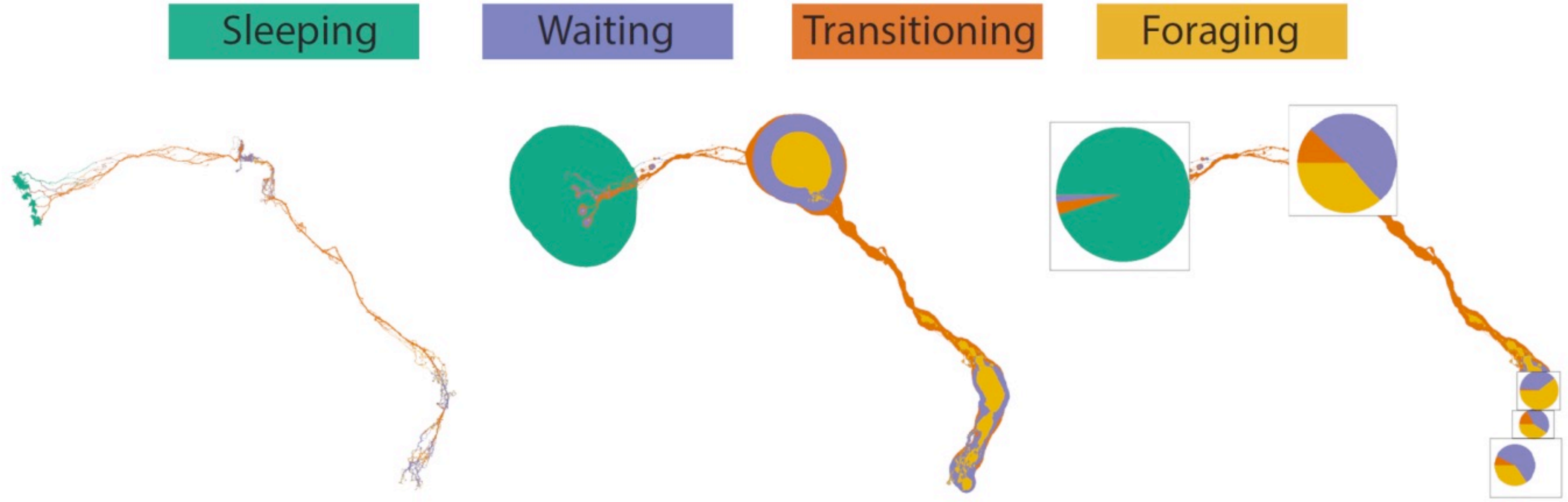




Eagereyes: https://eagereyes.org/techniques/pie-charts

Is it Always bad to use Pie Charts?

PixelWise Pie Charts (Janetzko & Stein)



(a) original movement data (b) circular point placement (c) pixel-based pie charts

2018. “Pixel Wise Pie Charts: Placement of Data Points Visualizing Spatial Value Distributions. Janetzko & Stein

VisGuides: Visualization Forum

VisGuides

discourses.

Pie Charts are bad and 3D Pie Charts are very bad.

Visual Design   Pie Charts

Peter   Jul 23

Hi, everybody,

As a data analyst in a large organization, I read about how bad is to use pie charts and in particular 3D pie charts. However, the managers whom I work for really like pie charts, especially 3D ones. I usually label each wedge with the actual value and wonder if this is good enough to alleviate the problems raised in this guideline. These managers are not that stupid. Their instinct may tell them or us something. I thus wonder if pie charts have some merits that may explain why many people like these.

Source: https://www.stevefenton.co.uk/2009/04/pie-charts-are-bad/ 4

http://www.businessinsider.com/pie-charts-are-the-worst-2013-6?IR=T 3

created Jul 23 | last reply Aug 5 | 2 replies | 114 views | 3 users | 4 links

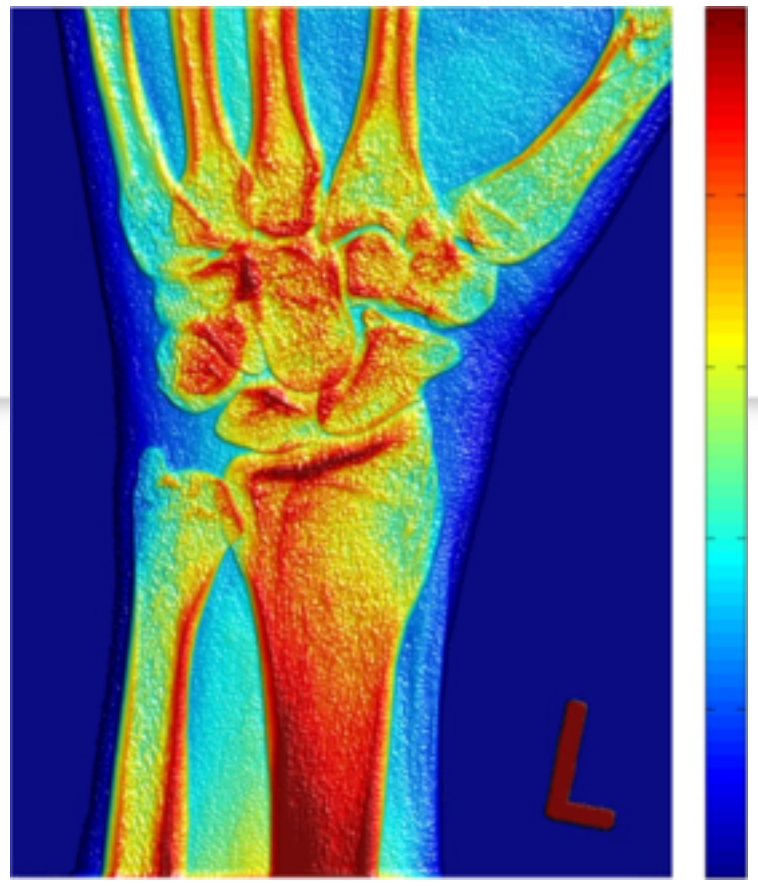
http://visguides.org/

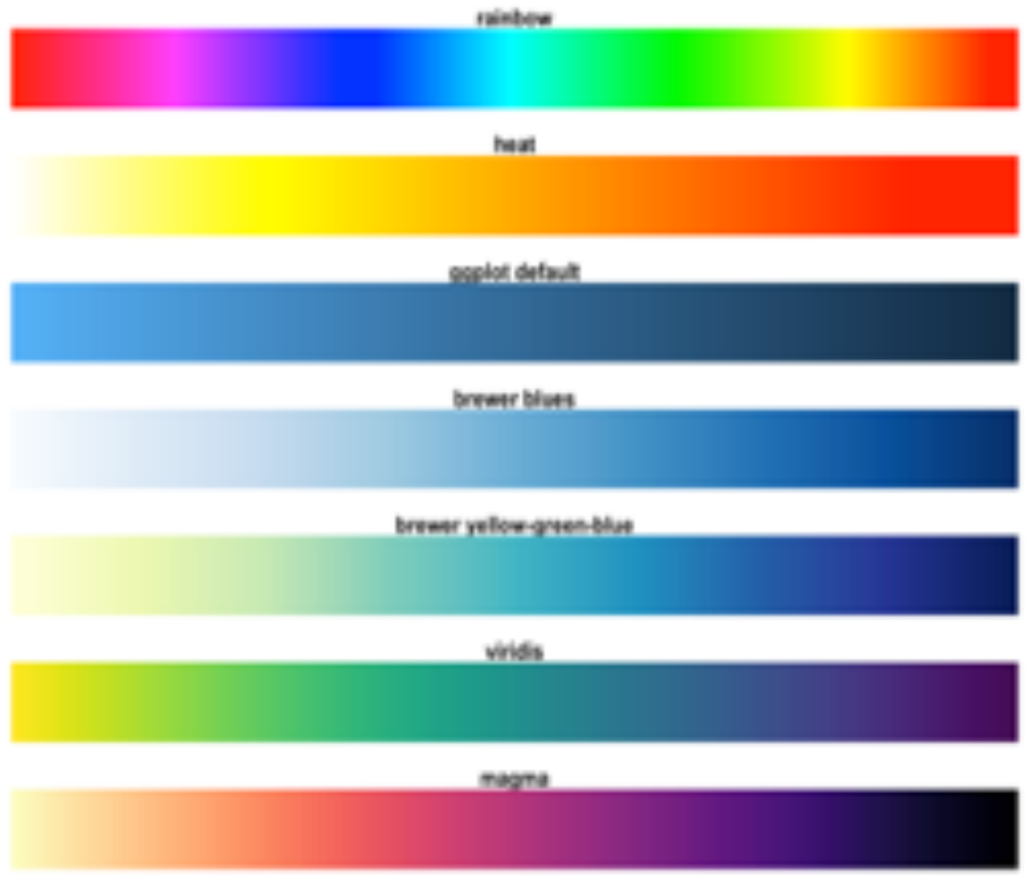
Top Discussed Guidelines

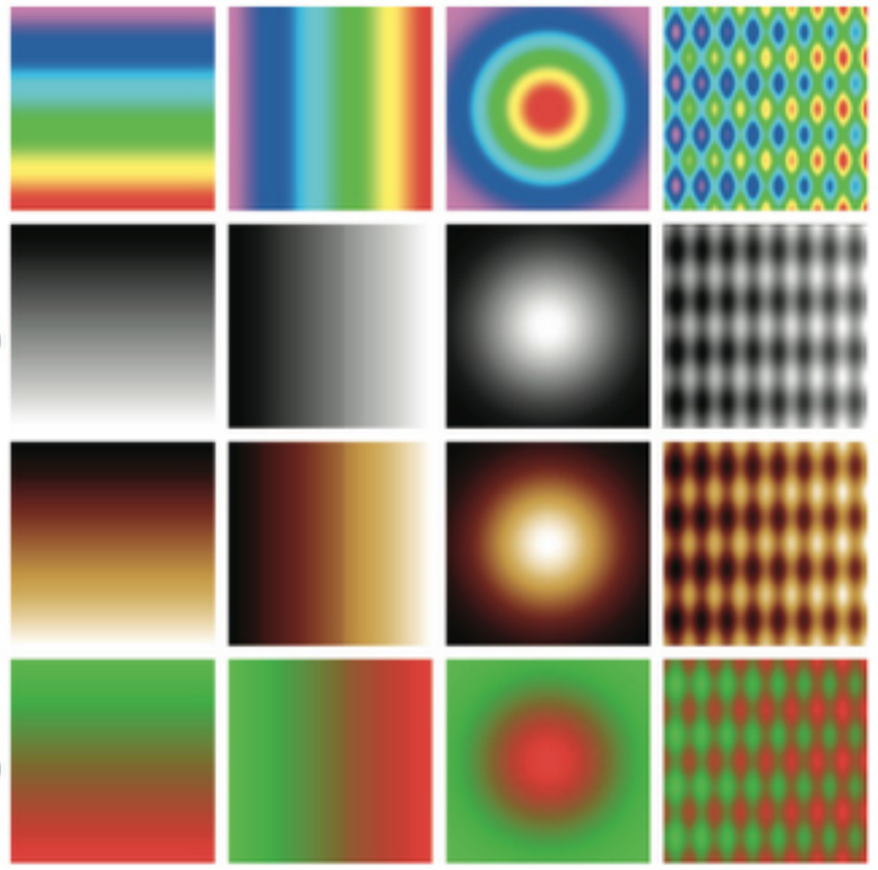
Topic	Replies	Views	Activity
Data-Ink Ratio Principle, How to use it? Theory	7	10.2k	26d
What are the main disadvantages of 3D visualizations in general? Visual Design	4	7.3k	May 2018
Use of gendered colours in visualization: a guideline or a personal principle? Color	5	6.5k	26d
Dimensionality of Scatter Plot: 1D? 2D? Theory	2	4.4k	26d
Tools for checking visualization colors for color blinds Color	5	3.7k	Jan 2021
(Don't / Do) Replicate the Real World in VR? VR/VE	8	3.4k	Apr 2019
Action without Interaction Interaction	5	3.4k	24d

Rainbow Color Map is considered Harmful

VisGuides







Rainbow Colormap

Perception Color

enviroscientist

Oct '17

Guideline: Rainbow color Map is considered harmful
Source: D Borland and M T Russell II, Rainbow Color Map (Still) Considered Harmful, CG&A, 27(2), 2007.

Question:
I represent a group of environmental scientists. We see and create visualizations with rainbow colormaps in thousands. It would be a pain if everyone uses a different colormap for each variable in these visualization. Is there a standard colormap we can use as a default map that everyone understand? Can visualization researchers be more constructive by recommending a colormap that maximize the perceptual bandwidth while minimize the problems such as being unsuitable for color blindness?

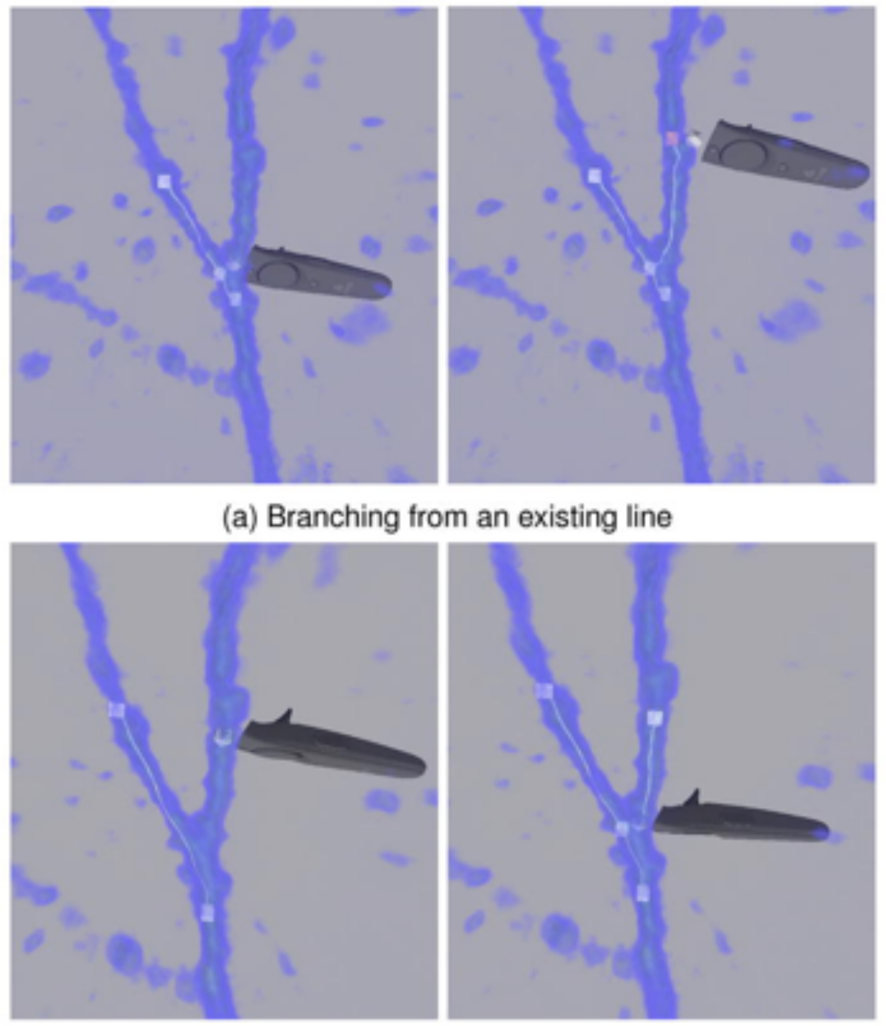
Don’t Replicate the Real World

(Don’t / Do) Replicate the Real Worl

Visual Design

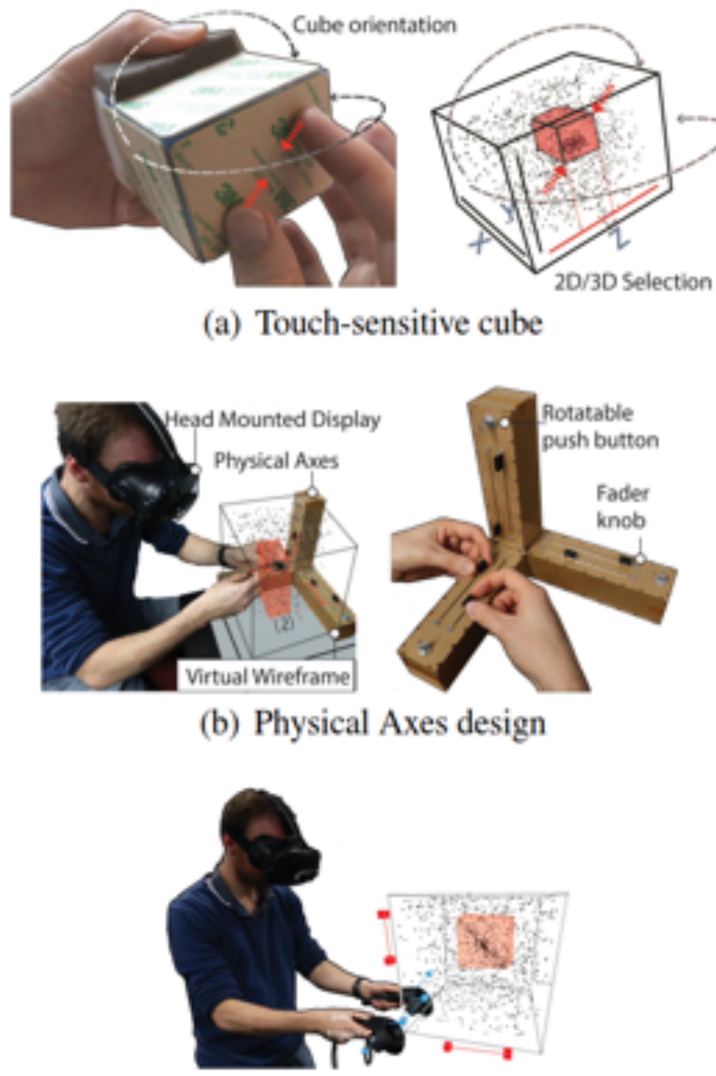
matthias.kraus

Guideline: Don’t Replicate the Real World
Source: https://sites.umiacs.umd.edu/elm/2017/10/03/3d-vi guide challenges/ 6



(a) Branching from an existing line

(b) Connecting a branch back to the parent tree



(a) Touch-sensitive cube

(b) Physical Axes design

(c) Virtual mid-air design

Question:
The overall message of the guideline given in the above blog is clear (Rule Nr. 7): For 3D visualizations for nonspatial data, it is unfavorable to replicate real-world controls. However, this seems to only cover 3D visualizations on the screen and not fully immersive visualization environments in VR. I think that in such environments, at least for some application cases, the exact opposite would be helpful: try to replica the real world as far as possible and extend/improve/adapt it to smooth away difficulties given by real environments (e.g., limitations given by screen sizes: the entire wall of the virtual environment could be used for display). This would probably help the analyst to quickly orient in and interact naturally with the virtual environment without a steep learning curve. Of course, the new technology enables new ways to interact with the environment and we should exploit those possibilities, but I also think that it would increase the overall usability to create a familiar environment to the known real world and therefore use a “normal” room as the “base” of the virtual environment and implement visual metaphors.
What is your opinion on that?

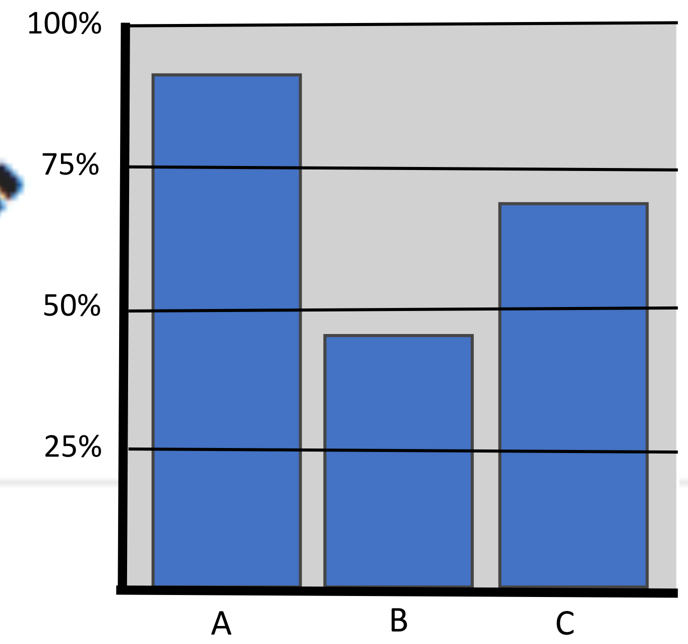
Good Visualisation should Maximise Data-ink Ratio

Data-Ink Ratio Principle, How to use it?

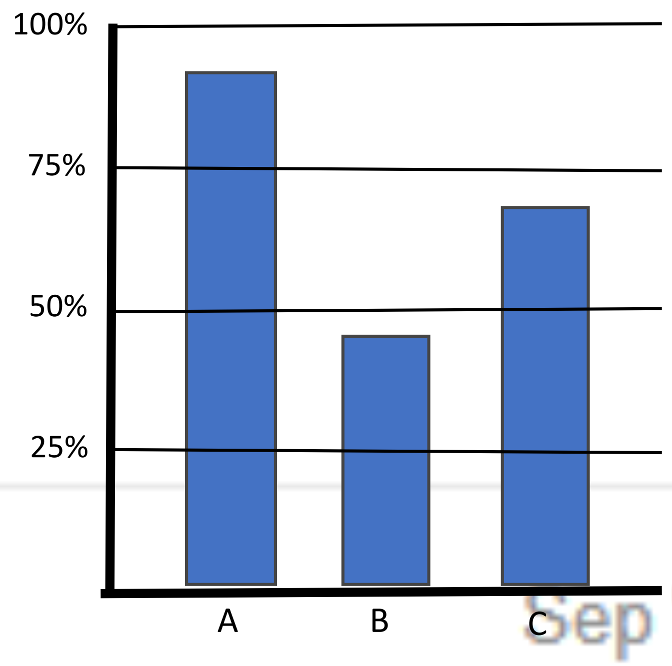
Theory

sureweb

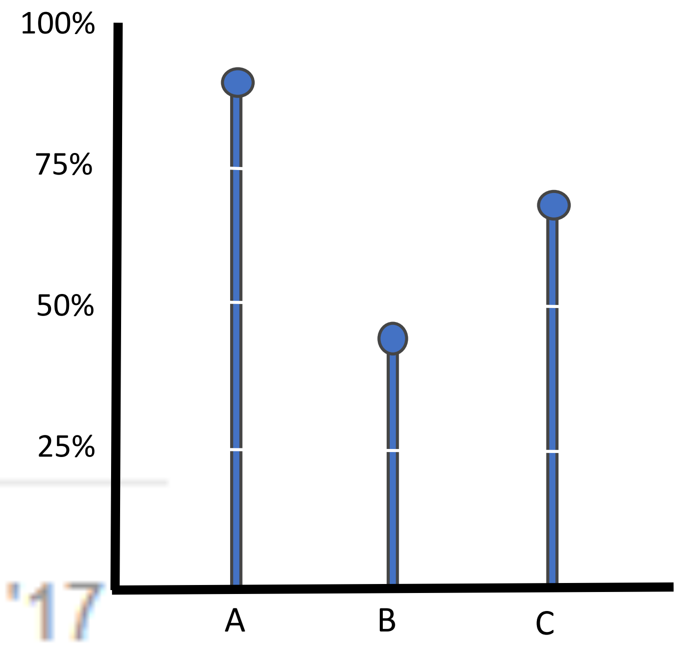
Sep '17



Low Data-Ink Ratio



Higher Data-Ink Ratio



Highest Data-Ink Ratio

Guideline: Good visualization should maximize data-ink ratio
Source: Edward R Tufte, The Visual Display of Quantitative Information, Graphics Press, 1983.

I have some difficulties to follow this guideline and need some advice. On a computer screen, white colors use more energy. In a bar chart, if I use a black or colored background, do I use less non-data-ink than the white background? If I use wider bars, do I use more data-ink than narrow bars? If I use a dot to represent the top of each bar, so I use less data-ink than bars?

Reply

Further Reading

Diehl et al.